

## House Price Prediction – Project Summary

This project aimed to predict house prices using machine learning techniques based on property features such as area, number of bedrooms, bathrooms, parking availability, and furnishing status. The dataset was explored, cleaned, and prepared for model development.

The housing dataset contained 545 records and 13 columns. Data quality checks showed no significant missing values, and categorical features were converted into numerical format using one-hot encoding. The dataset was then divided into training and testing sets using an 80/20 split.

Two regression models were developed: Linear Regression and Random Forest Regressor. Their performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and  $R^2$  Score. The Linear Regression model achieved the best overall performance, explaining approximately 65% of the variation in house prices.

Feature analysis indicated that area, number of bathrooms, parking spaces, air conditioning, and preferred location were among the most influential factors affecting house prices. Larger properties with additional amenities generally commanded higher market values.

The project demonstrates that machine learning can effectively support real estate price estimation. Real estate businesses can use these insights to improve pricing strategies, focus marketing efforts on high-value property features, and make more informed investment decisions.